Artificial Intelligence

Course Code: CS-202

Introduction to Artificial Intelligence and its History

Instructor: Habiba Arshad

Course administration

- **Instructor:** Habiba Arshad (Lecturer, Deptt of Computer Science, University of Wah, Wah Cantt).

- **Contact Information:** habiba.arshad@uow.edu.pk

For queries related to course(Assignments/Projects) just leave an email

at **habiba.arshad@uow.edu.pk**

during office timings 08:00AM - 04:00 PM

Evaluation Criteria (2+1 Credit hrs)

Assessment	%Age
Assignments/ Quizzes	10% + 15% = 25%
Mid Term Exam	30%
Final Term Exam	45%
Total	100

Assignments	Quizzes
▪ Assignments will be due at the beginning of the class. Under normal circumstances, late work will not be accepted. ▪ Students are expected to do their own assignments. ▪ Plagiarism will be observed strictly.	• Oral quizzes to check the understanding of previous lecture before starting the next. • Announced quizzes to assess the understanding of taught topics.

Few Things to Remember

- Show sensibility in your conduct.
- Learning should be your primary objective.
- Attendance will be marked within 10 minutes at the start of the class.
- 80% of your attendance is mandatory.
- **Zero tolerance policy** on attendance, discipline in class during lectures.
- Assignments must be submitted on time, no late submissions.
- In case of copied assignment both parties will be given **zero.**
- Don’t miss your Classes, Quizzes, Presentations, Assignments and Projects/Presentations.
- You all need to be a good human being.

Google Classroom

z4xwa5yb

Course Objective

The key objectives of this course include:

- To comprehend the fundamentals of Artificial Intelligence
- To evaluate and implement an appropriate informed or uninformed search algorithm for a problem and characterize its time and space complexity
- To translate natural language sentences (e.g., English) into logical statements
- To convert logic statements into a clause form and apply resolution to a set of logic statements to answer a query

Objective: Course Learning Outcomes (CLOs)

Discuss

- **CLO1:** Discuss the key concepts in the field of artificial intelligence

Implement

- **CLO2:** Implement artificial intelligence techniques and case Studies

Model

- **CLO3:** Model an automated solution for given problems using artificial intelligence paradigm

Graduate Attributes

GA's/CLOs	CLO1	CLO2	CLO3
GA1 (Academic Education)			
GA2 (Knowledge for Solving Computing Problems)	✓		
GA3 (Problem Analysis)		✓	
GA4 (Design/Development of Solutions)			✓
GA5 (Modern Tool Usage)			
GA6 (Individual and Teamwork)			
GA7 (Communication)			
GA8 (Computing Professionalism and Society)			
GA9 (Ethics)			
GA10 (Life-long Learning)			

Course Information

Text Books

- Artificial Intelligence A Modern Approach by Stuart Russell and Peter Norvig, 4th edition, 2020

Reference Book

- Paradigms of Artificial Intelligence Programming: Case studies in Common Lisp”, by Norvig, P.,original edition, 1992.
- AI algorithms, data structures, and idioms in Prolog, Lisp, and Java by Luger, G.F. and Stubblefield, W.A., 6 th Edition, 2009.

Course Overview

- An Introduction to Artificial Intelligence and its Applications towards Knowledge-Based Systems
- Intelligent Agents and Problem solving Agents
- Basic algorithms for Problem Solving by Searching
- Game-playing and Adversarial Search Case Studies
- Expert Systems
- Logic and Reasoning
- Knowledge Representation
- Learning from examples;
- Natural Language Processing
- Fuzzy logics and Machine Learning

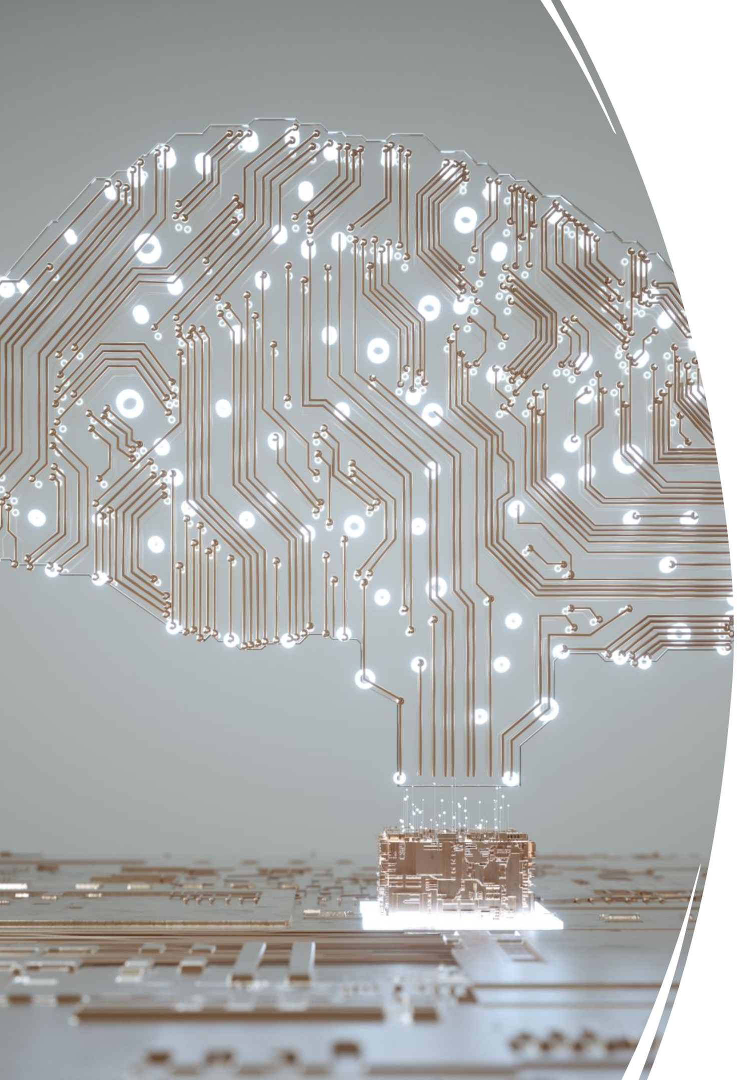
OBE

CLO Covered in this Lecture:

CLO1: **Discuss** the key concepts in the field of artificial intelligence

Why Study AI?

- Studying Artificial Intelligence (AI) is important because it helps us design machines and systems that can think, learn, and make decisions like humans.
- AI is used in many areas such as healthcare, education, business, robotics, and everyday applications like voice assistants and recommendation systems.
- It improves efficiency, automates tasks, and solves complex real-world problems.
- Learning AI also opens doors to exciting career opportunities and helps us understand both technology and human intelligence better, while encouraging us to use it responsibly for the benefit of society.



What is Artificial Intelligence?

Artificial Intelligence

- **Artificial Intelligence (AI)** is a branch of computer science focused on creating systems capable of performing tasks that normally require **human intelligence**.
- These tasks include:
  - Understanding natural language
  - Recognizing patterns
  - Making decisions
  - Learning from experience
- AI combines ideas, methods, and knowledge from multiple academic disciplines to solve complex problems and create intelligent behavior in machines.

History and Evolution of Artificial Intelligence

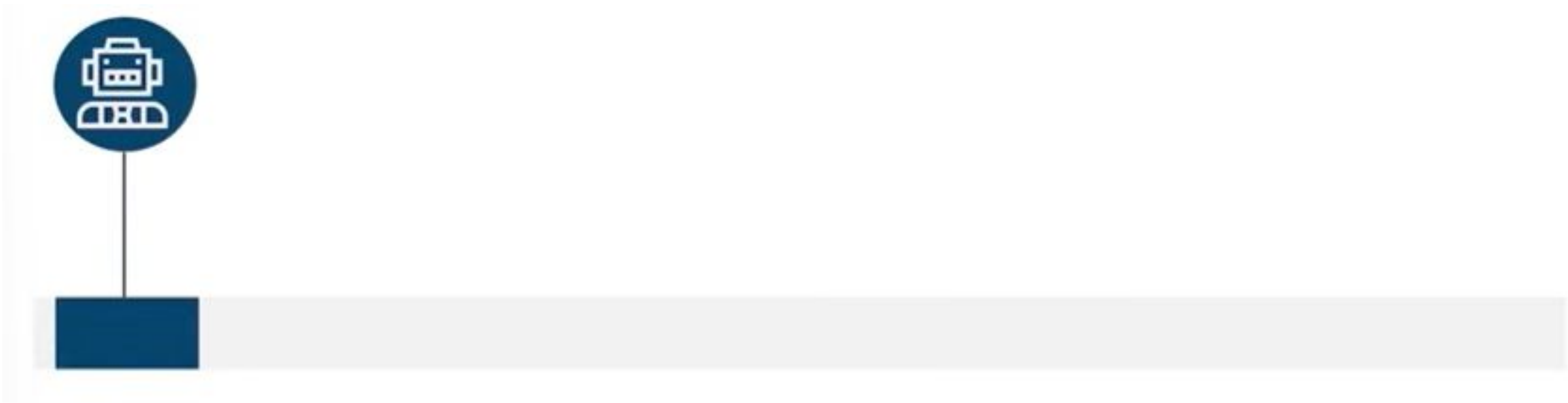
The concept of **Artificial Intelligence** actually goes back to **ancient times**.

History and Evolution of Artificial Intelligence

Greek mythology-Talos

Talos was a giant animated bronze warrior programmed to guard the island of **Crete,** created by Hephaestus.

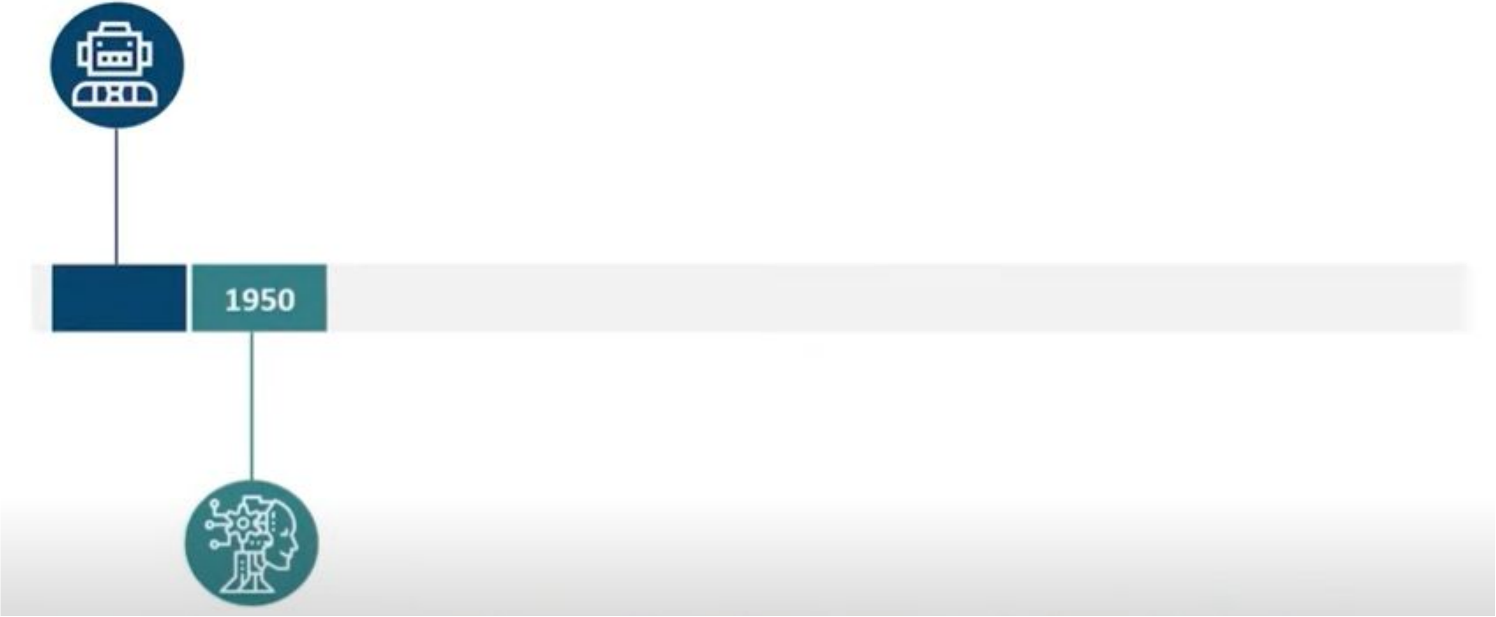
Talos was considered an *automaton* — an early vision of what we now call a robot — programmed to follow commands and defend the island.



History and Evolution of Artificial Intelligence

1950-Alan Turing

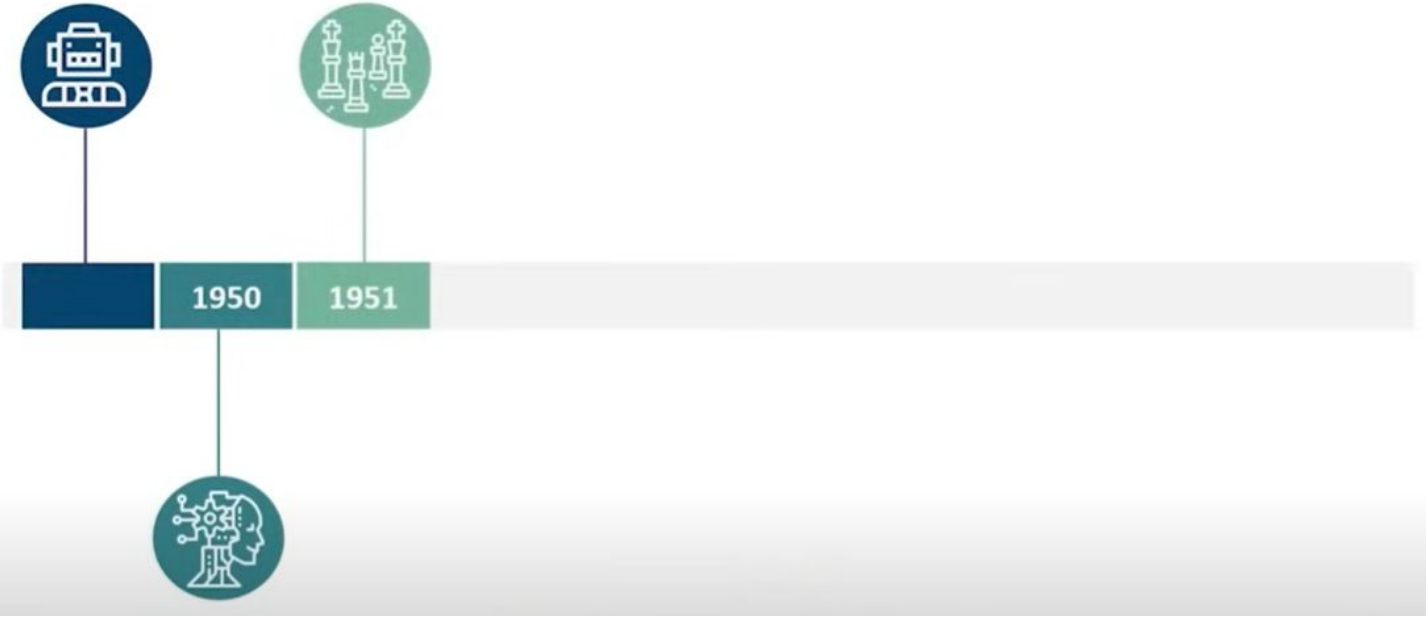
Alan Turing proposed the **Turing Test**, a method to determine whether a machine can think and respond like a human.
The Turing Test was the **first serious framework** in the *philosophy of Artificial Intelligence* and remains a foundational concept in AI research.



History and Evolution of Artificial Intelligence

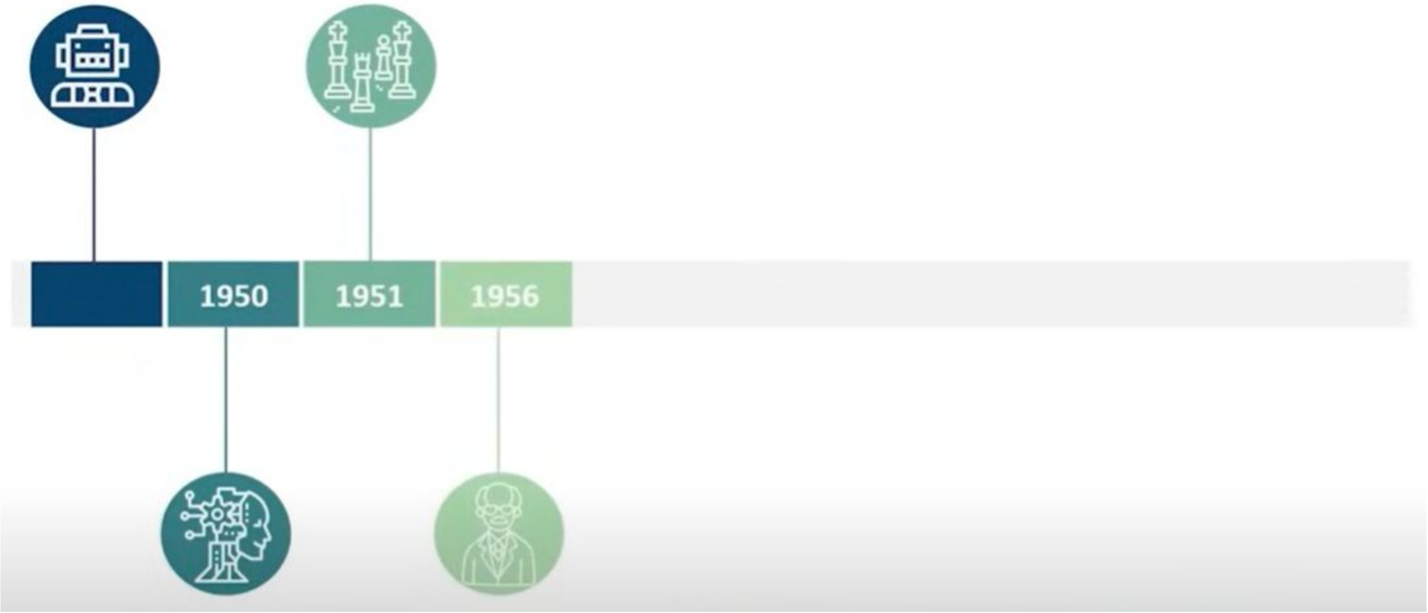
1951 – This era marked the beginning of **Game AI**, when computer scientists developed programs for **checkers** and **chess**.

Christopher Strachey wrote a checkers program, and Dietrich Prinz wrote for chess program.
These early programs laid the groundwork for modern AI in games and decision-making systems.



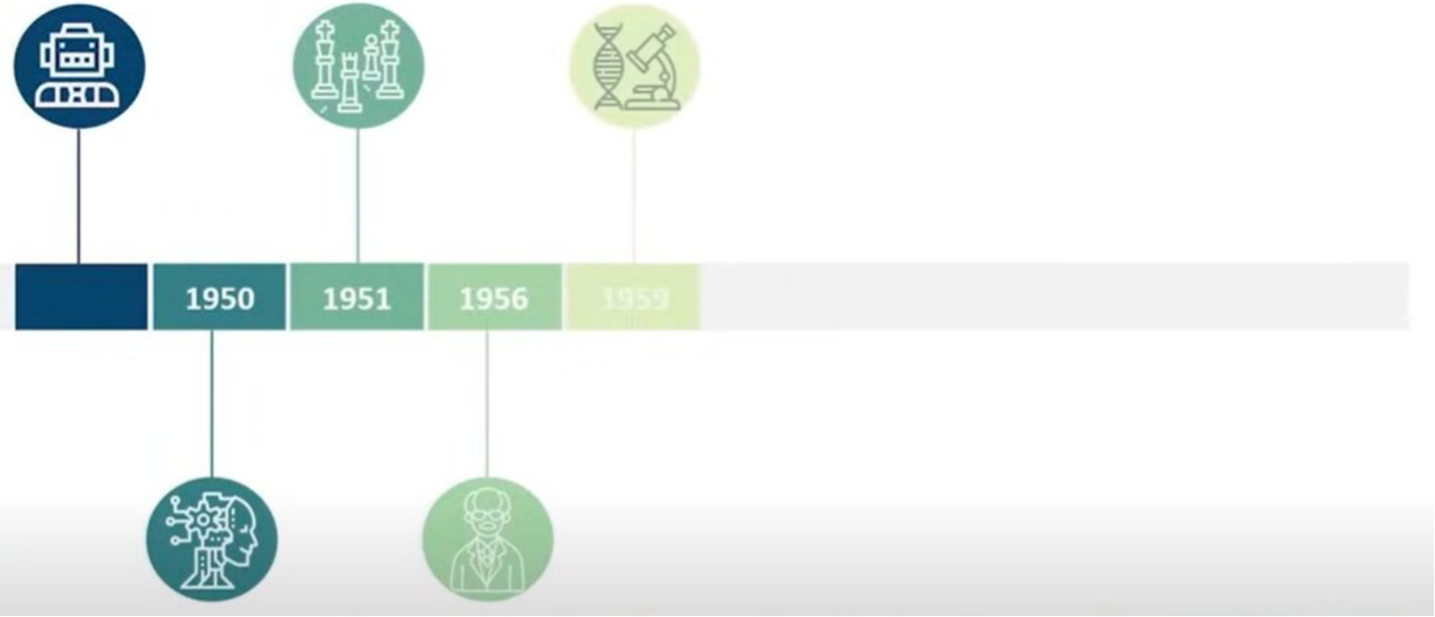
History and Evolution of Artificial Intelligence

1956 – A historic year for AI!
John McCarthy coined the term **“Artificial Intelligence”** during the **Dartmouth Conference**, which is considered the **birth of AI as a field of study**.



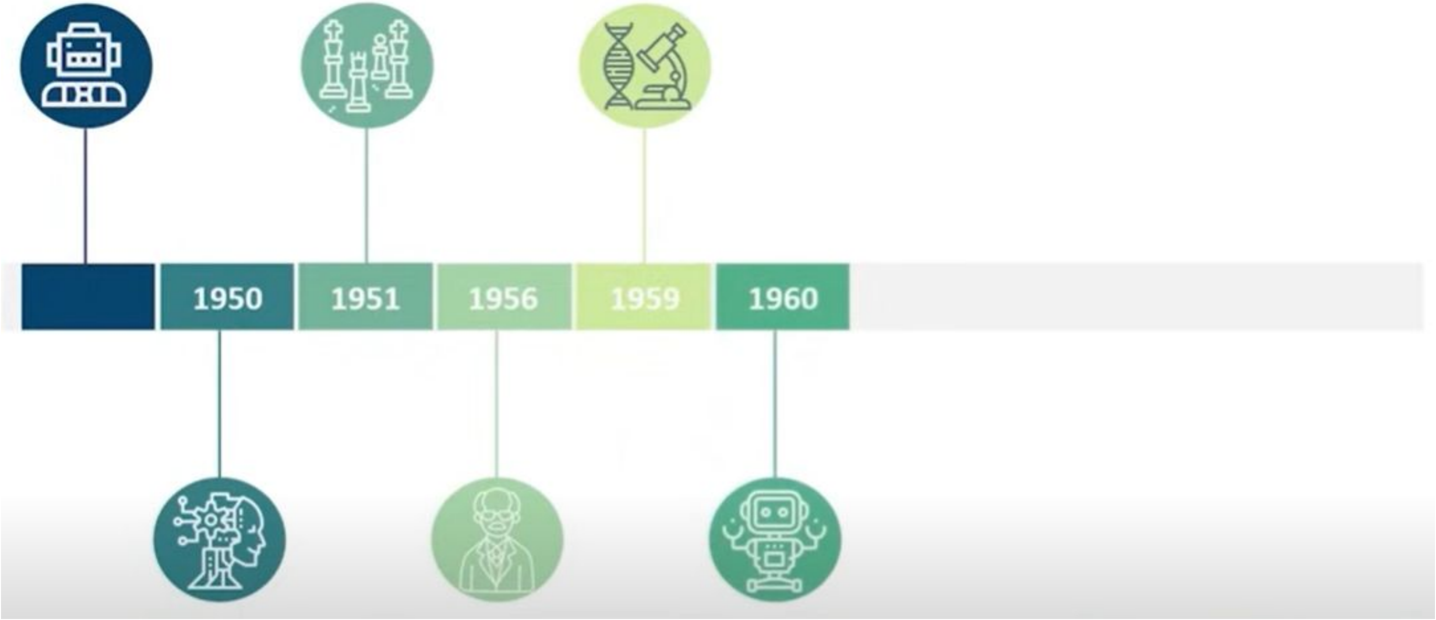
History and Evolution of Artificial Intelligence

1959 – The first **AI Laboratory** was established at **MIT**.
The **MIT AI Lab** became a major center for AI research and innovation.



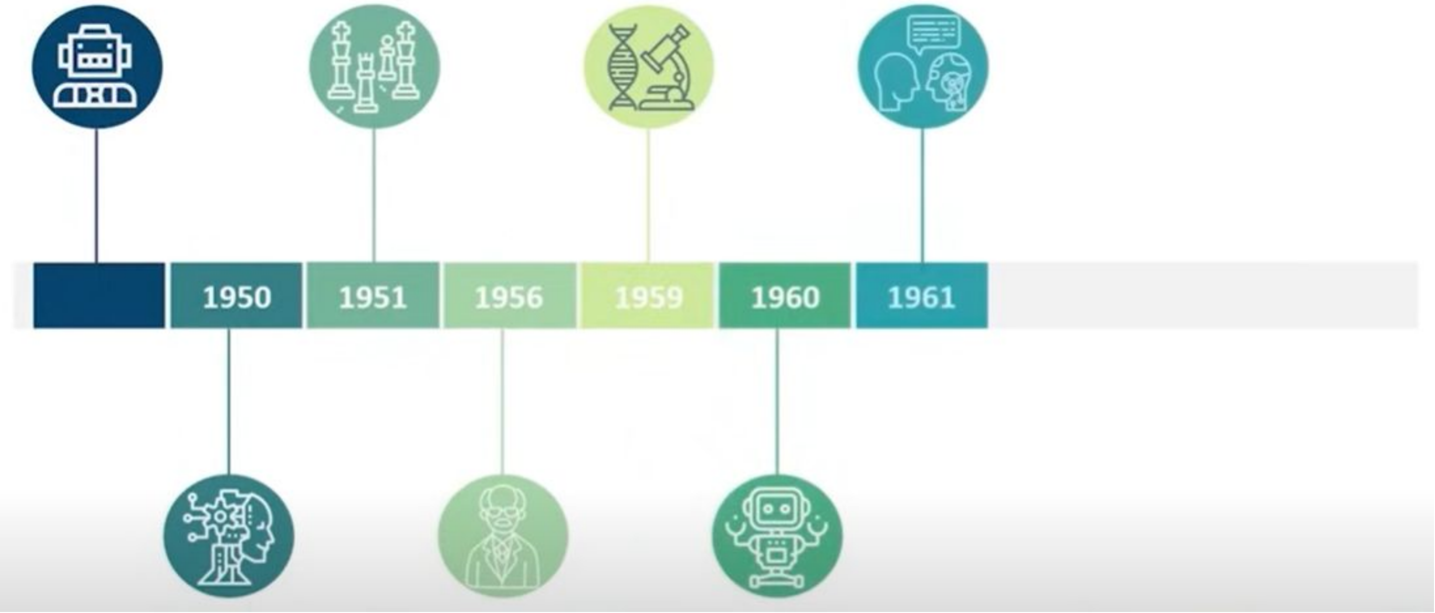
History and Evolution of Artificial Intelligence

1960 – The first **industrial robot** was introduced to the **General Motors assembly line**, marking the start of automation in manufacturing.



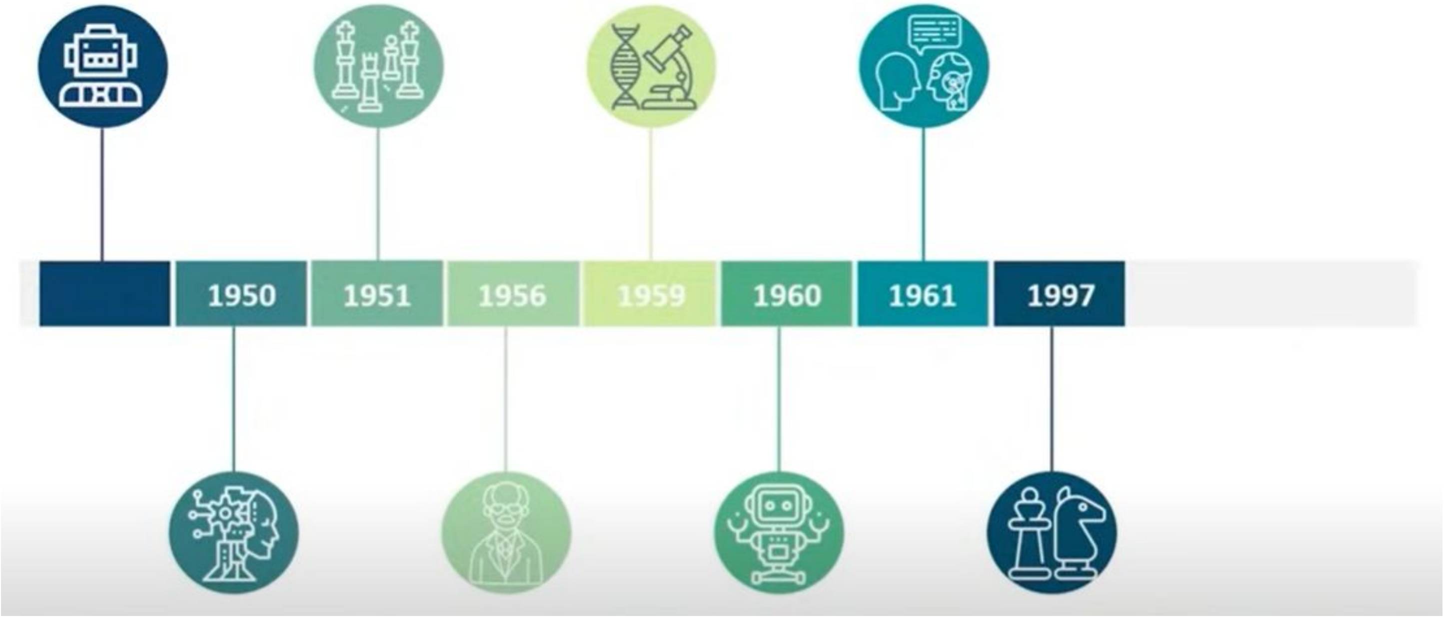
History and Evolution of Artificial Intelligence

1961 – The first **AI chatbot**, named **ELIZA**, was created by Joseph Weizenbaum.
ELIZA could simulate a conversation with a human — an early example of natural language processing.



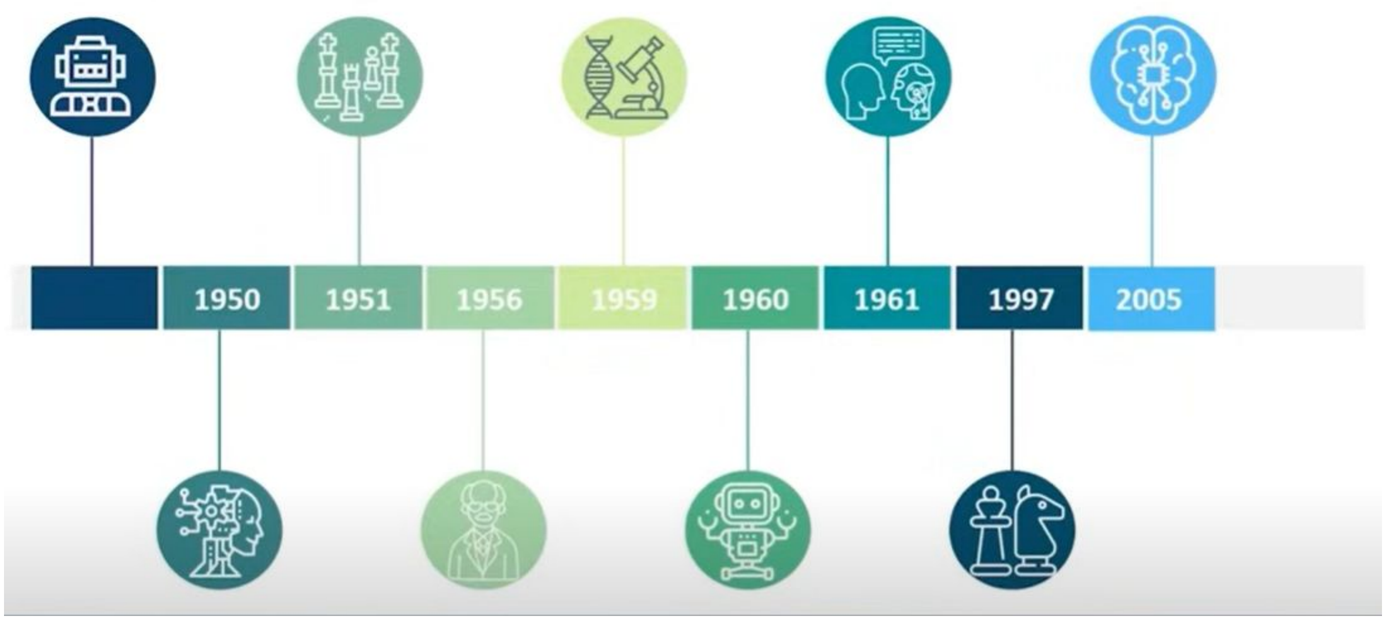
History and Evolution of Artificial Intelligence

1997 – IBM's Deep Blue made history by defeating world chess champion **Garry Kasparov**, showcasing AI's ability to outperform humans in complex strategy games.



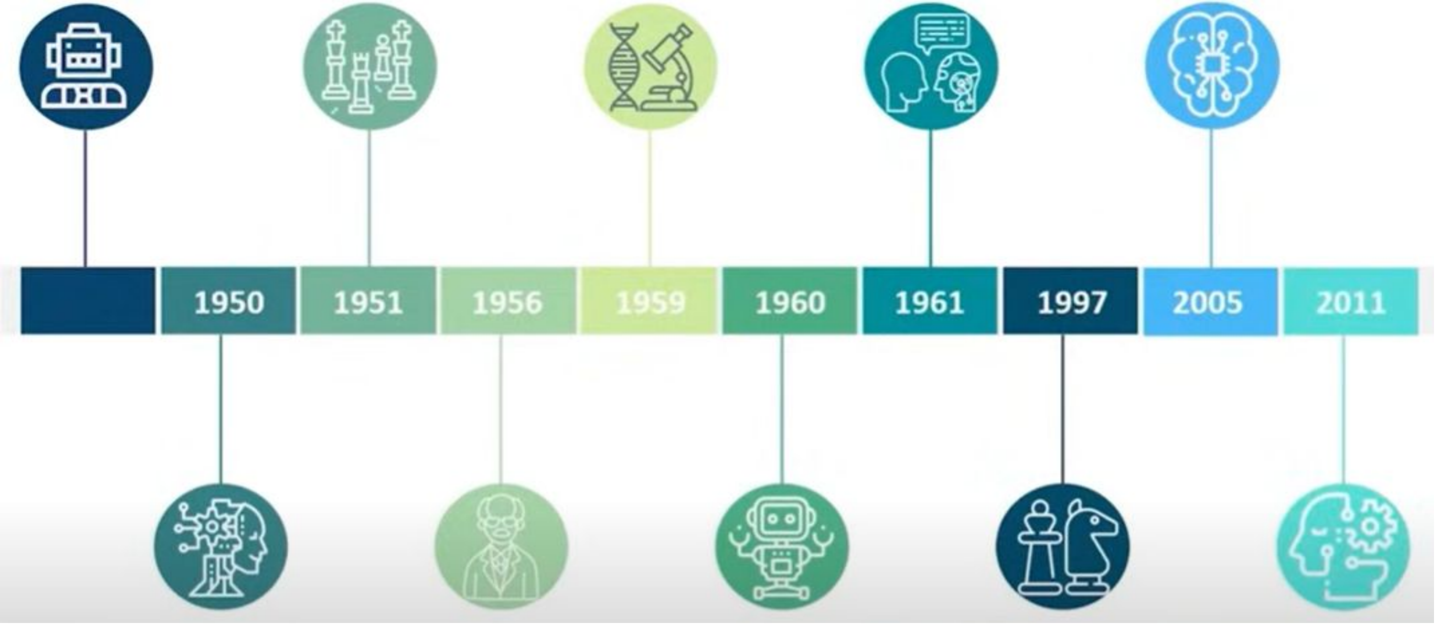
History and Evolution of Artificial Intelligence

2005 – An autonomous robotic car named **Stanley** won the **DARPA Grand Challenge**, proving that AI could navigate real-world environments independently.



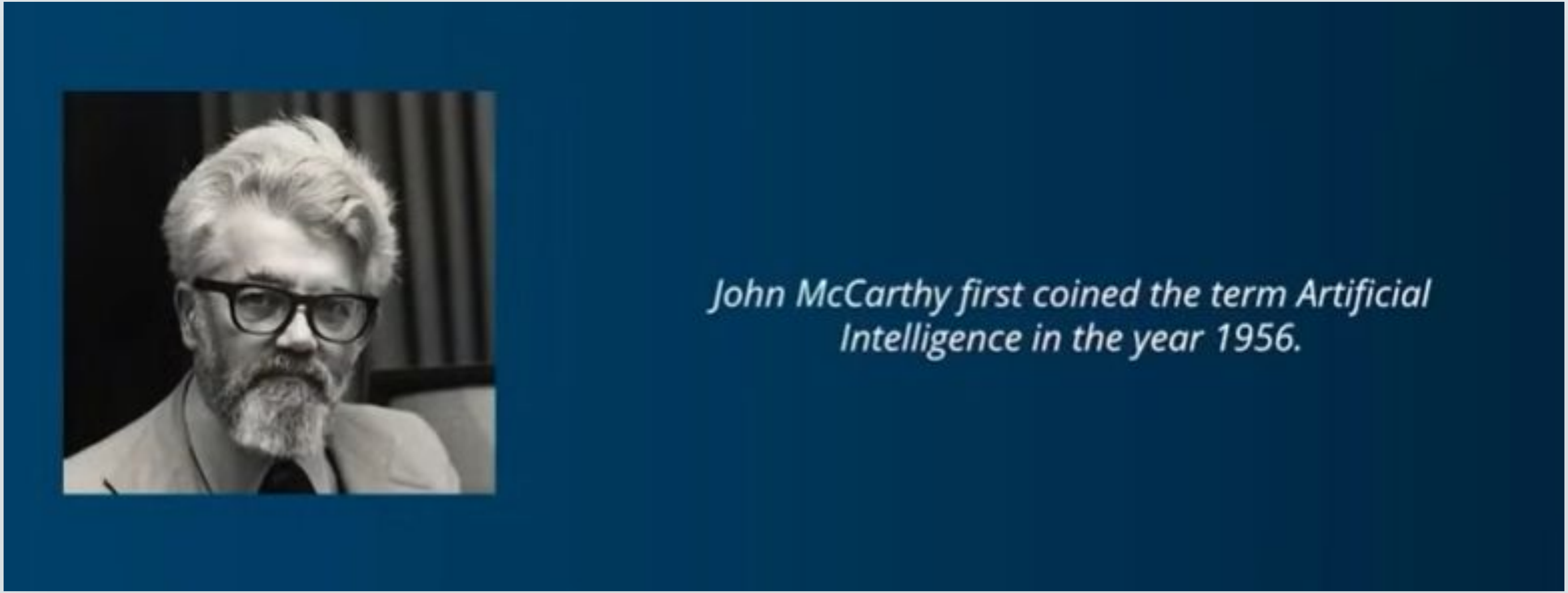
History and Evolution of Artificial Intelligence

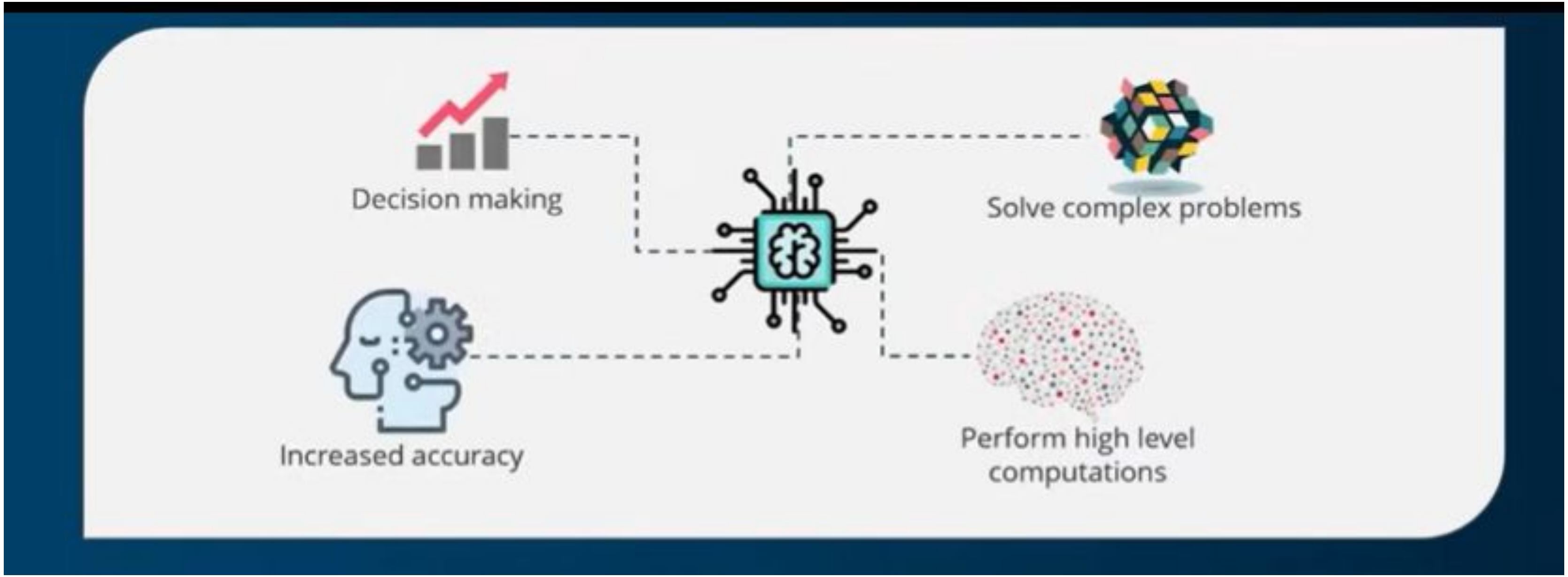
2011 – IBM's Watson, a question-answering AI system, defeated Jeopardy champions **Brad Rutter** and **Ken Jennings**, demonstrating AI's ability to understand and process human language at a deep level.



What is Artificial Intelligence?

AI as the science and engineering of making intelligent machines





AI can also be defined as:

“The development of computer systems that are capable of performing tasks that require human intelligence such as decision making, object detection, solving complex problems, and so on.”